An Erdős–Sós-type theorem for antidirected trees

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Abstract

We prove that every digraph D with more than $(t-2)|V(D)|$ arcs contains every antidirected tree on $t \geq 2$ vertices, establishing the conjecture of Addario-Berry, Havet, Linhares Sales, Reed, and Thomassé. The proof adapts the permutation-counting argument of the recent Erdős–Sós proof obtained in the FrontierMath Erdős benchmark. The result also gives Sumner's tournament bound for antidirected trees. An accompanying Lean 4 formalization proves the density theorem.

Acknowledgments

ChatGPT 6 Astra was used to develop the argument, write the exposition, and produce the accompanying Lean formalization. The author's role was minimal, consisting mostly of asking questions towards trying to understand the recent Erdős–Sós proof and its possible extensions.

1 Introduction

An *oriented tree* is obtained by assigning a direction to every edge of a tree. It is *antidirected* if every vertex is a source or a sink; equivalently, no vertex has both positive indegree and positive outdegree. We prove an analogue of the Erdős–Sós theorem for these trees in digraphs.

Throughout the paper, a digraph is finite and loopless, but it may contain both opposite arcs between a pair of vertices. There is at most one arc in each ordered direction. Copies are injective and need not be induced, and every target tree has $t \geq 2$ vertices. We write $e(D)$ for the number of arcs of D.

We prove the following density statement, conjectured by Addario-Berry, Havet, Linhares Sales, Reed, and Thomassé in 2013 [1].

Theorem 1.1. *Let T be an antidirected tree on $t \geq 2$ vertices. Every digraph D satisfying*

$$e(D) > (t-2)|V(D)|$$

contains a copy of T.

1.1 Related work

Several special cases were known. Addario-Berry et al. proved the conjecture for antidirected trees of diameter at most three [1]. Stein and Trujillo-Negrete later proved it for every antidirected caterpillar, and for every antidirected tree when the host excludes three specified orientations of $K_{2,\lceil (t-1)/12 \rceil}$ [3]. Stein and Zárate-Guerén also obtained an asymptotic version for balanced antidirected trees of bounded maximum degree and linear order in large oriented graphs [5].

1.2 Connection with the Erdős–Sós proof

Our proof is based on the new proof of the Erdős–Sós conjecture obtained in the FrontierMath Erdős benchmark and recorded in [6].[1] We adapt its counting of marked vertex permutations and induction on rooted trees; we do not claim an independent origin for this method. The directed argument distinguishes arcs leaving the first vertex of a permutation from arcs entering it. We check that the two operations in the graph proof, adding a leaf and joining branches at a root, respect the required arc directions. The graph theorem itself is not used as a black box.

Conversely, our theorem includes the Erdős–Sós assertion for graphs as a special case. Given a graph G, form the symmetric digraph $\overleftrightarrow{G}$ by replacing each edge with its two opposite orientations. Given a tree U, choose its bipartition and orient every edge from one class to the other, obtaining an antidirected tree T. Then

$$e(\overleftrightarrow{G}) = 2e(G), \qquad \overleftrightarrow{G} \text{ contains } T \quad\Longleftrightarrow\quad G \text{ contains } U.$$

Thus Theorem 1.1, applied to $\overleftrightarrow{G}$, specializes precisely to the Erdős–Sós assertion that $2e(G) > (t-2)|V(G)|$ forces a copy of U.

1.3 Outline of the proof

For each permutation $(v_1, \ldots, v_N)$ of the host vertices, we mark position $i \geq 2$ according to whether $v_1 \to v_i$ or $v_i \to v_1$. We count pairs consisting of a permutation and a marked position for which the first i vertices contain a copy of the target tree, with its root mapped to v_1. Each pair is counted once, regardless of how many such copies it contains.

The induction has two cases. If the root is a leaf, we start with a copy of the smaller tree lying before the current marked position. The vertex at that position is then available for the new leaf. If the root is not a leaf, we split the tree into two smaller trees meeting only at the root. Reordering the permutation gives a counting inequality in which the two copies use disjoint vertices outside that root. The direction of the marking arc is unchanged when branches are joined, and is reversed relative to the root when a leaf becomes the new root. In an antidirected tree, neighbors have opposite roles as sources and sinks, while branches at a vertex share the same role. These facts allow the induction to proceed.

Section 2 defines the marked pairs and computes their number. Section 3 gives a separate proof for antidirected paths, illustrating the sign-switching reversal. Section 4 proves the two counting inequalities for rooted trees, and Section 5 completes the induction. The tournament consequence and the Lean formalization are given in Sections 6 and 7.

2 Marked permutations

Let D be a digraph on $N \geq 1$ vertices, and let

$$\pi = (v_1, v_2, \ldots, v_N)$$

be an ordering of its vertices. A position $i \geq 2$ is *positive* if $v_1 \to v_i$ and *negative* if $v_i \to v_1$. Write

$$\mathcal{O}_D^+ = \{(\pi, i) : 2 \leq i \leq N,\ v_1 \to v_i\},$$
$$\mathcal{O}_D^- = \{(\pi, i) : 2 \leq i \leq N,\ v_i \to v_1\},$$

[1]The repository is maintained by Tom Adamczewski. Its provenance account credits the graph proof to GPT-6 Astra, working autonomously in the FrontierMath Erdős benchmark of Adamczewski and Bloom.

and put $M^+ = |\mathcal{O}_D^+|$ and $M^- = |\mathcal{O}_D^-|$. A position may have both signs when both opposite arcs are present, and it has neither sign when the two vertices are nonadjacent.

Proposition 2.1 (Basic counts). *For every digraph D on $N \geq 1$ vertices,*

$$M^+ = M^- = e(D)(N-1)!. \tag{2.1}$$

Proof. If the first vertex is x, every ordering of the remaining vertices has $d_D^+(x)$ positive positions. Hence

$$M^+ = (N-1)! \sum_{x \in V(D)} d_D^+(x) = e(D)(N-1)!.$$

The negative count is identical, using indegrees instead of outdegrees. □

For a pair (π, i), its *prefix* is the vertex set $\{v_1, \ldots, v_i\}$. Let (H, h) be a rooted oriented graph. For $\varepsilon \in \{+, -\}$, define $\mathcal{O}_D^\varepsilon(H, h)$ to be the set of pairs in $\mathcal{O}_D^\varepsilon$ whose prefix contains a copy of H mapping h to v_1. We say that such a pair *supports* (H, h). The copy need not follow the order of the permutation, and each pair is counted only once. When the host is fixed, we write $\mathcal{O}^\varepsilon(H, h)$. The notation $-\varepsilon$ denotes the sign opposite to ε.

3 Warm-up: antidirected paths

The density bound in Theorem 3.3 is already known: it is a special case of the antidirected-caterpillar theorem of Stein and Trujillo-Negrete [3]. We give a separate permutation-counting proof to illustrate the sign-switching mechanism.

For paths rooted at an endpoint, we need only one operation: extend an earlier path by the vertex at the current marked position and make that vertex the new root. The root changes from a source to a sink, or conversely. Prefix reversal changes the marking sign in exactly the same way. We prove the path case separately to isolate this mechanism before introducing branch joining.

For $k \geq 1$ and $\varepsilon \in \{+, -\}$, let P_k^ε be the antidirected path with k arcs, rooted at an endpoint that is a source if $\varepsilon = +$ and a sink if $\varepsilon = -$. Thus the vertex sequence starts $x_0 \to x_1 \leftarrow x_2 \to \cdots$ in the positive case and $x_0 \leftarrow x_1 \to x_2 \leftarrow \cdots$ in the negative case. Let A_k^ε count the marked states of sign ε whose prefixes contain a copy of P_k^ε rooted at the first vertex. These are existence counts, not counts of embeddings. Every mark of sign ε supplies the required one-arc path, so

$$A_1^\varepsilon = M^\varepsilon = e(D)(N-1)!. \tag{3.1}$$

Lemma 3.1 (Extending an antidirected path). *For every $k \geq 1$ and $\varepsilon \in \{+, -\}$,*

$$A_k^\varepsilon \leq A_{k+1}^{-\varepsilon} + N!. \tag{3.2}$$

Proof. For each ordering contributing to A_k^ε, discard its first supporting marked state (π, i_0). There are at most $N!$ discarded states. At any remaining state (π, i), the earlier prefix contains P_k^ε rooted at v_1 and avoids v_i. If $\varepsilon = +$, the marking arc is $v_1 \to v_i$; adjoining it extends the path with a new sink endpoint v_i. If $\varepsilon = -$, the marking arc is $v_i \to v_1$ and the new endpoint is a source. In both cases the extended path, rooted at v_i, is $P_{k+1}^{-\varepsilon}$.

Reverse the prefix through position i:

$$(\pi, i) \longmapsto \big((v_i, v_{i-1}, \ldots, v_1, v_{i+1}, \ldots, v_N), i\big). \tag{3.3}$$

The prefix vertex set is unchanged, the first vertex is now v_i, and the marking arc has sign $-\varepsilon$ relative to that vertex. The image is therefore counted by $A_{k+1}^{-\varepsilon}$. The index i is retained and a second reversal recovers the source, so the map is injective. Adding back the discarded states proves (3.2). □

The reversal changes the ordering, not the directions of any arcs. Nor must the supported path follow the permutation order: only its root and its containing prefix are prescribed.

Example 3.2 (A source endpoint becomes a sink)**.** Let D have vertices a, b, c, d and arcs

$$a \to b, \qquad c \to b, \qquad a \to c, \qquad a \to d.$$

For $\pi = (a, b, c, d)$, position 3 is a positive mark supporting the two-arc path $a \to b \leftarrow c$ rooted at a. The marking arc $a \to c$ is not part of that path. At the later positive mark 4, the earlier copy avoids d, so adjoining $a \to d$ gives

$$d \leftarrow a \to b \leftarrow c.$$

Reversal sends $(\pi, 4)$ to $((d, c, b, a), 4)$. The new state is negative, with marking arc $a \to d$, and supports the displayed three-arc path rooted at its sink endpoint d.

Theorem 3.3 (Antidirected-path density bound)**.** *Let P be any antidirected path on $t \geq 2$ vertices. If a digraph D contains no copy of P, then*

$$e(D) \leq (t-2)|V(D)|.$$

Proof. The empty host is immediate, so put $N = |V(D)| \geq 1$. We claim that for every $k \geq 1$ and each sign ε,

$$e(D)(N-1)! \leq A_k^{\varepsilon} + (k-1)N!. \tag{3.4}$$

The case $k = 1$ is (3.1). For $k > 1$, apply the inductive inequality with sign $-\varepsilon$, followed by Lemma 3.1, to obtain

$$e(D)(N-1)! \leq A_{k-1}^{-\varepsilon} + (k-2)N! \leq A_k^{\varepsilon} + (k-1)N!.$$

This proves the claim. Choose an endpoint of P and let ε be its source or sink sign. Since D is P-free, $A_{t-1}^{\varepsilon} = 0$. Taking $k = t-1$ in (3.4) and dividing by $(N-1)!$ proves the bound. $\square$

The coefficient is sharp: a complete symmetric digraph on $t-1$ vertices has $(t-1)(t-2)$ arcs and cannot contain P. The same sign-switching reversal will add a leaf to an arbitrary rooted oriented tree in Lemma 4.2. The extra operation needed for general antidirected trees is joining branches at a common root, which we develop next.

4 Joining copies and adding a leaf

We first compare the counts for two trees that meet at a single root. This step does not require the trees to be antidirected. That assumption enters when we use both counting inequalities in the induction.

Lemma 4.1 (Joining at a root)**.** *Let T_1 and T_2 be rooted oriented subtrees of an oriented tree T, all rooted at r, such that*

$$T_1 \cup T_2 = T, \qquad V(T_1) \cap V(T_2) = \{r\}.$$

Then, for either sign ε,

$$|\mathcal{O}^{\varepsilon}(T_1, r)| + |\mathcal{O}^{\varepsilon}(T_2, r)| \leq M^{\varepsilon} + N! + |\mathcal{O}^{\varepsilon}(T, r)|. \tag{4.1}$$

Proof. We also allow the pairs

$$\mathcal{Z} = \{(\pi, 1) : \pi \text{ is an ordering of } V(D)\}, \qquad |\mathcal{Z}| = N!.$$

We construct an injection

$$\mathcal{O}^{\varepsilon}(T_1, r) \setminus \mathcal{O}^{\varepsilon}(T, r) \longrightarrow \big(\mathcal{O}^{\varepsilon}_D \setminus \mathcal{O}^{\varepsilon}(T_2, r)\big) \cup \mathcal{Z}.$$

Take a pair (π, i) in the domain and let $a \leq i$ be its first position of sign ε supporting (T_1, r). Write

$$\pi = (x, A, B, C), \qquad A = (v_2, \ldots, v_a), \quad B = (v_{a+1}, \ldots, v_i), \quad x = v_1.$$

Here A, B, C are consecutive lists of vertices; the original prefix ends after B. Set $b = 1 + |B|$ and send the pair to

$$((x, A, B, C), i) \longmapsto ((x, B, A, C), b). \tag{4.2}$$

If B is empty, the image belongs to $\mathcal{Z}$. Otherwise its first vertex is still x and the last vertex of its prefix is still v_i, so its marking arc has the same sign ε.

The vertices in x, A support a copy of T_1 rooted at x. If the image prefix x, B supported T_2 rooted at x, the two copies would be disjoint outside x and could be joined to embed T in the original prefix. This contradicts the source condition. Thus the image belongs to the stated set.

For the inverse, read B from the image prefix after x. Starting immediately after that prefix, recover A as the shortest nonempty initial list whose last vertex supplies a mark of sign ε at x and which, together with x, supports T_1 rooted at x. The original A satisfies these conditions, and no proper initial segment does, by the minimality of a. This test does not include B. The remaining entries are C, and the original index is $i = 1 + |A| + |B|$. This also recovers the source when $b = 1$. The map is therefore injective. Since support of T implies support of T_1, taking cardinalities yields

$$|\mathcal{O}^{\varepsilon}(T_1, r)| - |\mathcal{O}^{\varepsilon}(T, r)| \leq M^{\varepsilon} - |\mathcal{O}^{\varepsilon}(T_2, r)| + N!,$$

which is (4.1). □

We next add a leaf and make it the new root, using the same construction as in Lemma 3.1, now for an arbitrary rooted oriented tree. Reversing the prefix interchanges the first and marked vertices. The marking arc itself does not change, but its sign does: an arc entering the old root leaves the new root, and conversely.

Lemma 4.2 (Adding a leaf)**.** *Let (S, p) be a rooted oriented tree. Add a new leaf ℓ adjacent to p, producing T, and root T at ℓ. Define*

$$\sigma = \begin{cases} +, & \ell \to p, \\ -, & p \to \ell. \end{cases}$$

Then

$$|\mathcal{O}^{-\sigma}(S, p)| \leq |\mathcal{O}^{\sigma}(T, \ell)| + N!. \tag{4.3}$$

Proof. Let $\mathcal{Z}$ again be the $N!$ pairs $(\pi, 1)$. We construct an injection

$$\Psi : \mathcal{O}^{-\sigma}(S, p) \longrightarrow \mathcal{O}^{\sigma}(T, \ell) \cup \mathcal{Z}.$$

For each ordering π, send its first pair in $\mathcal{O}^{-\sigma}(S, p)$, if one exists, to $(\pi, 1) \in \mathcal{Z}$.

For any later pair (π, i), an earlier prefix contains S rooted at v_1, and therefore does not use v_i. Reverse the first i entries:

$$(\pi, i) \longmapsto ((v_i, v_{i-1}, \dots, v_1, v_{i+1}, \dots, v_N), i). \tag{4.4}$$

If $\sigma = +$, then the source is negative, so $v_i \to v_1$; this arc represents $\ell \to p$ and is positive after the reversal. If $\sigma = -$, then the source is positive, so $v_1 \to v_i$; this arc represents $p \to \ell$ and is negative after the reversal. In either case the image supports (T, ℓ) and has sign σ. Prefix reversal is an involution for each fixed i, and the images in $\mathcal{Z}$ are distinct from one another and from the images with $i \geq 2$, so the map is injective. $\square$

5 Proof of the directed density theorem

For a vertex r of an antidirected tree T, put

$$\sigma_T(r) = \begin{cases} +, & r \text{ is a source,} \\ -, & r \text{ is a sink.} \end{cases}$$

Adjacent vertices have opposite signs. If we split a tree into branches at a root, the root remains a source in every branch or a sink in every branch. Thus Lemma 4.2 uses the appropriate sign at the new root, and Lemma 4.1 uses the same sign in both smaller trees.

Theorem 5.1 (The main counting inequality)**.** *Let T be an antidirected tree on $t \geq 2$ vertices, rooted at r, and let D be a digraph on $N \geq 1$ vertices. With $\sigma = \sigma_T(r)$,*

$$M^\sigma \leq |\mathcal{O}^\sigma(T, r)| + (t-2)N!. \tag{5.1}$$

Proof. We induct on t. If $t = 2$, the unique arc incident with r points away from r when $\sigma = +$ and towards r when $\sigma = -$. Every pair of sign σ therefore supplies the required rooted edge, so $M^\sigma = |\mathcal{O}^\sigma(T, r)|$.

Suppose first that r is a leaf, let p be its neighbour, and put $S = T - r$. The vertex p has sign $-\sigma$ in S. By induction and Proposition 2.1,

$$M^\sigma = M^{-\sigma} \leq |\mathcal{O}^{-\sigma}(S, p)| + (t-3)N!.$$

Lemma 4.2 now gives

$$M^\sigma \leq |\mathcal{O}^\sigma(T, r)| + (t-2)N!.$$

Suppose instead that r has degree at least two. Partition the components of $T - r$ into two nonempty groups, and let T_1, T_2 be the corresponding rooted subtrees after restoring r. Both have root sign σ. If $t_j = |V(T_j)|$, then $t_1 + t_2 = t + 1$. Adding the two induction inequalities and applying Lemma 4.1 gives

$$\begin{aligned} 2M^\sigma &\leq |\mathcal{O}^\sigma(T_1, r)| + |\mathcal{O}^\sigma(T_2, r)| + (t_1 + t_2 - 4)N! \\ &\leq M^\sigma + |\mathcal{O}^\sigma(T, r)| + (t-2)N!. \end{aligned}$$

Subtracting M^σ proves the result. $\square$

Proof of Theorem 1.1. The strict density hypothesis excludes the empty host. Write $N = |V(D)| \geq 1$, and root T at any vertex r. If D were T-free, then $\mathcal{O}^\sigma(T, r)$ would be empty. Theorem 5.1 and Proposition 2.1 would give

$$e(D)(N-1)! \leq (t-2)N!,$$

or equivalently $e(D) \leq (t-2)N$, contrary to the hypothesis. $\square$

6 A tournament consequence

Sumner's conjecture asserts that every tournament on $2t-2$ vertices contains every oriented tree on t vertices; it is known for all sufficiently large t [2, 4]. Theorem 1.1 gives the antidirected case for every $t \geq 2$.

Corollary 6.1 (Sumner's bound for antidirected trees). *Every tournament on at least $2t-2$ vertices contains every antidirected tree on $t \geq 2$ vertices.*

Proof. Restrict to any $N = 2t-2$ vertices. The resulting tournament D has

$$e(D) = \frac{N(N-1)}{2} = N\left(t - \frac{3}{2}\right) > (t-2)N.$$

Apply Theorem 1.1. □

7 Lean formalization

The accompanying project in `lean/` proves the counting inequalities and the density theorem in Lean 4.19.0, using the bundled `Std` library. We describe its definitions and their relation to the proof. Instructions for checking the files are given in `lean/README.md`.

7.1 Definitions and theorem statements

Finite digraphs are represented by lists of vertices and arcs, with no repeated entries. Opposite arcs are allowed. An embedding preserves arcs and is injective on all target vertices, not just separately on the sources and the sinks.

The final density theorem assumes that the target has at least two vertices, is antidirected, and has a connected underlying graph with no simple cycle. Connectedness is defined using walks, and a simple cycle has distinct vertices apart from its repeated endpoint. These definitions do not refer to the counting argument. The corresponding declarations are

- `Digraph.addarioBerry_bound_of_no_simple_cycle`, giving $e(D) \leq (t-2)|V(D)|$ for a T-free host;
- `Digraph.addarioBerry_contains_of_no_simple_cycle`, giving Theorem 1.1.

The older declarations `Digraph.addarioBerry_bound` and `Digraph.addarioBerry_contains` use a recursive definition of the target tree. Both versions remain available.

7.2 Relation to the proof

The counting proof is organized recursively. It starts with one arc, adds a new leaf as the root, or joins two trees at their common root. The code proves that the induction parameter is the number of target vertices minus two. It also proves that every antidirected tree with a connected, acyclic underlying graph can be built by these operations, with any chosen vertex as its root. This step is proved in `Digraph/TreeBridge.lean`; the final theorems do not require the user to supply a recursive construction.

The formal proof of Lemma 4.2 swaps the first and marked vertices rather than reversing the whole prefix. Both maps preserve the prefix vertex set, change the sign of the mark, and are their own inverses for each fixed marked position. The proof of Lemma 4.1 uses the same shortest marked prefix as the paper. The path density bound is a special case of the formalized general theorem; the separate warm-up proof has not been formalized line by line. The tournament corollary has not been formalized.

7.3 Verification

The project specifies the Lean version and includes a script that compiles the files and checks which axioms the theorems depend on. The theorem statements are collected in `lean/Statements.lean`. Only `propext`, `Classical.choice`, and `Quot.sound` are permitted in these dependency checks.

References

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